

Chapter 6

Supply, Demand and Government Policies

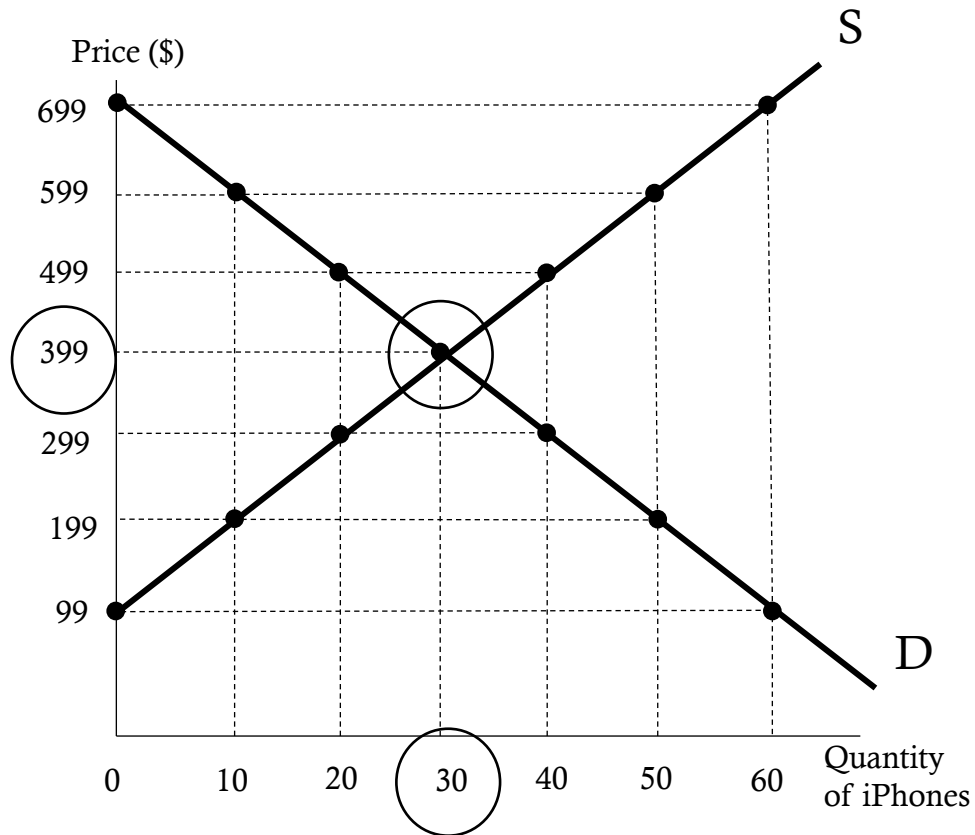
Learning Objectives

- LO 6.1: Identify and calculate the effects of government policies that place a ceiling on prices.
- LO 6.2: Identify and calculate the effects of government policies that put a floor under prices.
- LO 6.3: Illustrate and apply how a tax on a good affects the price of the good and the quantity sold.
- LO 6.4: Illustrate and apply how taxes levied on sellers and taxes levied on buyers are equivalent.
- LO 6.5: Recognize how the burden of a tax is split between buyers and sellers.

Key Points

- A price ceiling is a legal maximum on the price of a good or service. Rent control is an example. If the ceiling is below the equilibrium price, then it is binding, and the quantity demanded exceeds the quantity supplied. Because of the resulting shortage, sellers must somehow ration the good or service among buyers.
- A price floor is a legal minimum on the price of a good or service. The minimum wage is an example. If the price floor is above the equilibrium price, then it is binding, and the quantity supplied exceeds the quantity demanded. Because of the resulting surplus, buyers' demands for the good or service must somehow be rationed among sellers.
- When the government levies a tax on a good, the equilibrium quantity of the good falls. That is, a tax on a market shrinks the market's size.
- A tax on a good places a wedge between the price paid by buyers and the price received by sellers. When the market moves to the new equilibrium, buyers pay more for the good and sellers receive less for it. In this sense, buyers and sellers share the tax burden. The incidence of a tax (that is, the division of the tax burden) does not depend on whether the tax is levied on buyers or sellers.
- The incidence of a tax depends on the price elasticities of supply and demand. Most of the burden falls on the side of the market that is less elastic because that side of the market cannot respond as easily to the tax by changing the quantity bought or sold.

Market Equilibrium

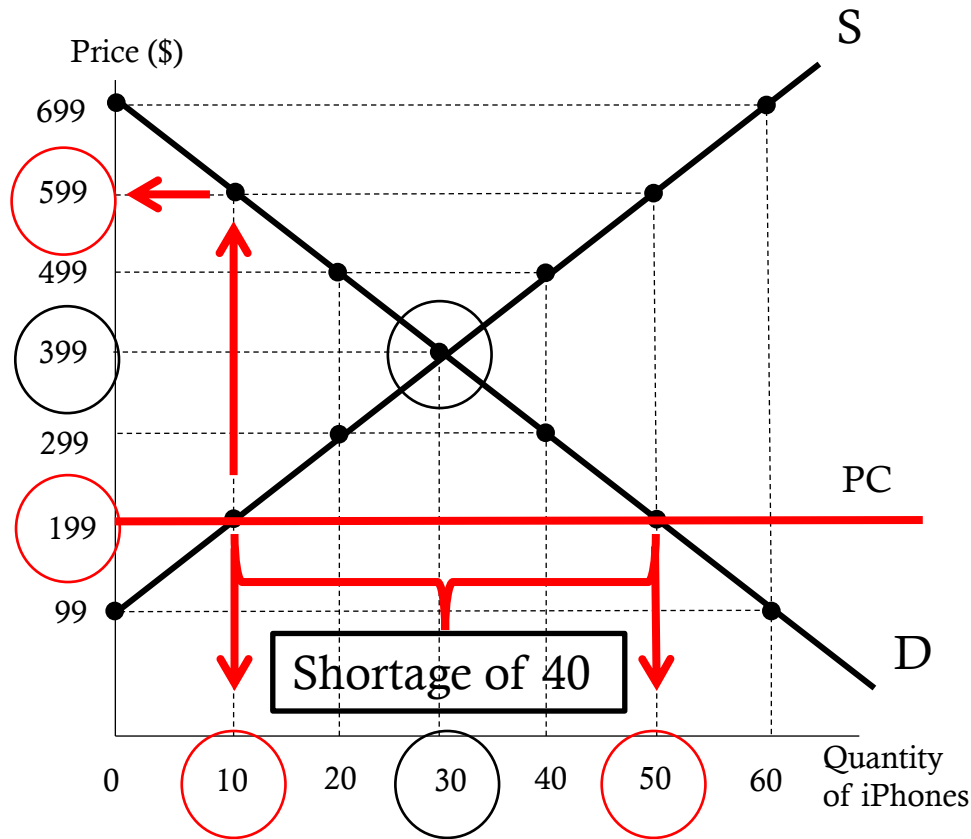


- **Recall:** Market equilibrium is a condition where there is no incentive for consumers or producers to alter their behavior.
- When one group alters their behavior a surplus or shortage arises, and eventually the market corrects itself to a new equilibrium.
- When pricing distortions are introduced into the market, the market cannot correct itself. These distortions are called *disequilibrium prices*.

Disequilibrium Prices

- Disequilibrium prices are often enacted by the government and come in two varieties:
- **Price ceilings** – a legal maximum price that can be charged for a good.
- For price ceilings to have an effect (be binding), they must be set below the market equilibrium price.
- A common example of price ceilings are rent controls. Goods such as gas, food, and auto parts have had their price restricted at some point.
- **Price floors** – a legal minimum price that can be charged for a good.
- For price floors to have an effect (be binding), they must be set above the market equilibrium price.
- A common example of a price floor is the minimum wage.

Price Ceiling



- Our market is in equilibrium at a price of \$399 and 30 iPhones.
- Suppose the government believes this to be an unfair price and sets a **price ceiling** (PC) of \$199.
- When a price ceiling is set no price above it can be charged.
- This result here is consumers want 50 iPhones and producers want to produce 10 iPhones.
- A shortage of 40 iPhones arises as a result of the price ceiling.
- **Note:** while the legal money price cannot rise above \$199, the actual price paid by the consumer (considering time spent waiting, black market premiums, etc.) is \$599.

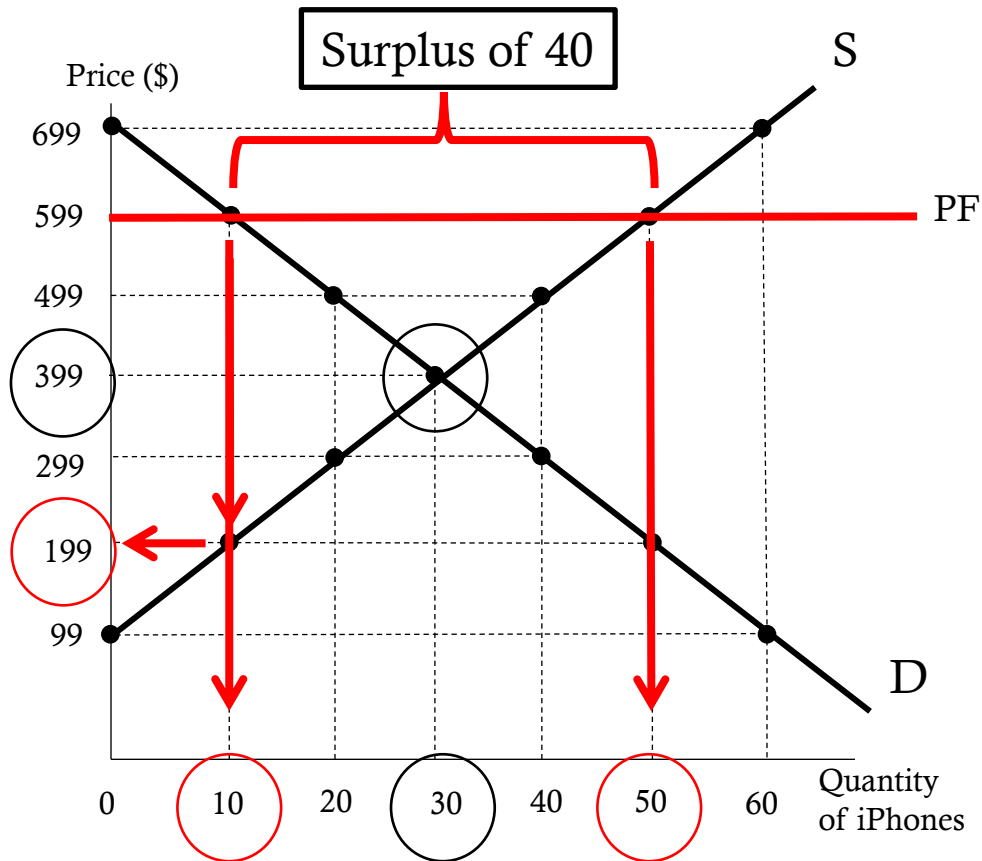
Activity – The Apartment Shortage

- The city of Metropolis sets a rent ceiling of **\$800/month**, below the market equilibrium of **\$1,200**.
- At \$800, **quantity demanded = 60,000 units, quantity supplied = 40,000 units**.
- What is the shortage?
- How might renters and landlords respond?
- Who gains, who loses, and what unintended consequences might emerge?
(*Think: time costs, quality decline, discrimination, black markets.*)

Activity – The Apartment Shortage

- Shortage = 20,000 units.
- Renters gain *lower prices* (if they find an apartment), but many are rationed out.
- Landlords reduce maintenance, convert units, or rent “off-book.”
- Market efficiency falls due to misallocation and deadweight loss.

Price Floors



- Our market is in equilibrium at a price of \$399 and 30 iPhones.
- Suppose the government believes this to be an unfair price for Apple and sets a **price floor** (PF) of \$599.
- When a price floor is set no price below it can be charged.
- This result here is consumers wanting 10 iPhones and producers want to produce 50 iPhones.
- A surplus of 40 iPhones arises as a result of the price ceiling.
- **Note:** while the legal money price cannot fall below \$599, the actual (real) price received by the producer (considering increased competition in trying to sell the product) is \$199.

Price Floors, Int'l Trade and Semiconductors

- In 1986, US semiconductor manufacturers were able to get a price floor established on Japanese semiconductors, mainly on dynamic random access memory (DRAM) chips.
- US semiconductor manufacturers sought protection because they thought the lower priced DRAM chips were harming their ability to compete, so they needed protection to buy time to retool their plants.
- In trade, and restriction will create a loser. The loser happened to be the US computer manufacturers and computer consumers.

“The agreement in fact has harmed American computer manufacturers, who have found themselves paying higher prices for computer chips. This makes American computer manufacturers less competitive and drives up computer prices for all Americans”

“As a result of the agreement, Japanese DRAM chips cost American computer makers 30 percent to 40 percent more than they cost European computer makers. Example: the European minimum price for a 1-meg DRAM chip is \$3.90; the price in the U.S. is \$5.00. The 4-meg chip costs \$18 in Europe but \$30 in the U.S.”

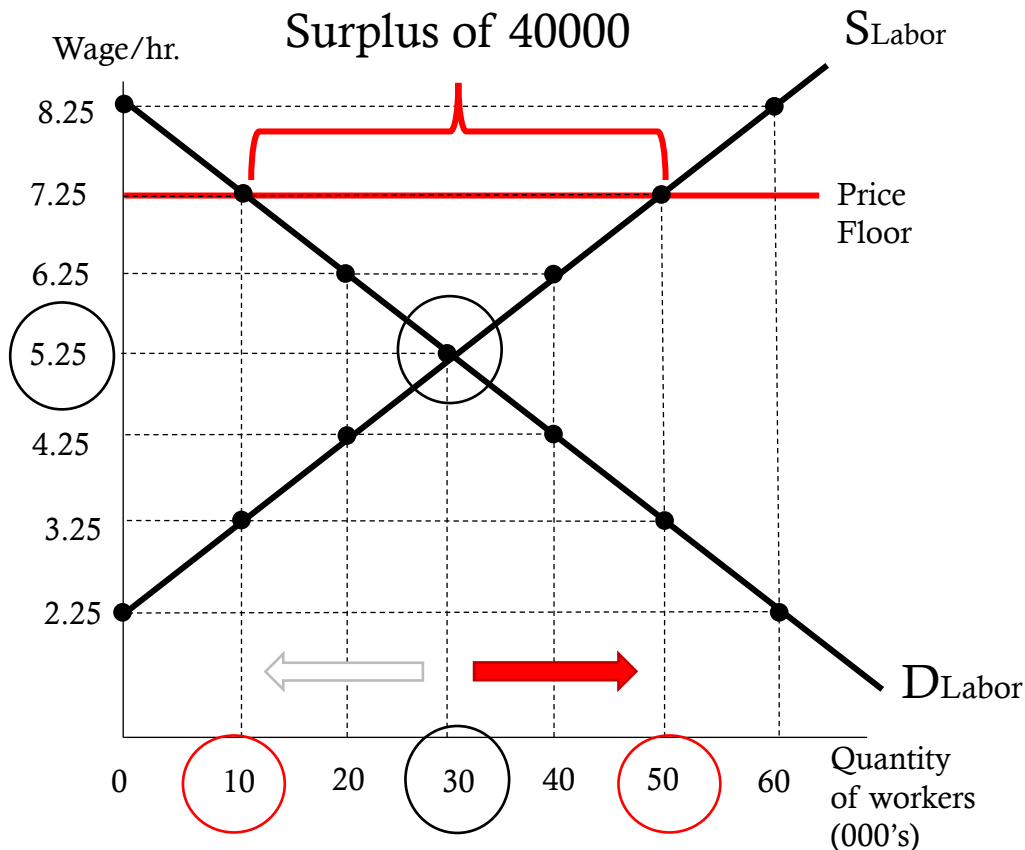
Activity – Raising the Wage Floor

- The minimum wage in Metroville rises from **\$10 to \$15/hour**.
- At \$10: firms **hire 100,000** workers; at \$15: firms **demand 70,000**.
- At \$15, **120,000** workers are willing to work.
- What is the labor surplus (unemployment)?
- Who benefits, and who bears the cost?
- How might employers and workers adjust behavior over time?

Activity – Raising the Wage Floor

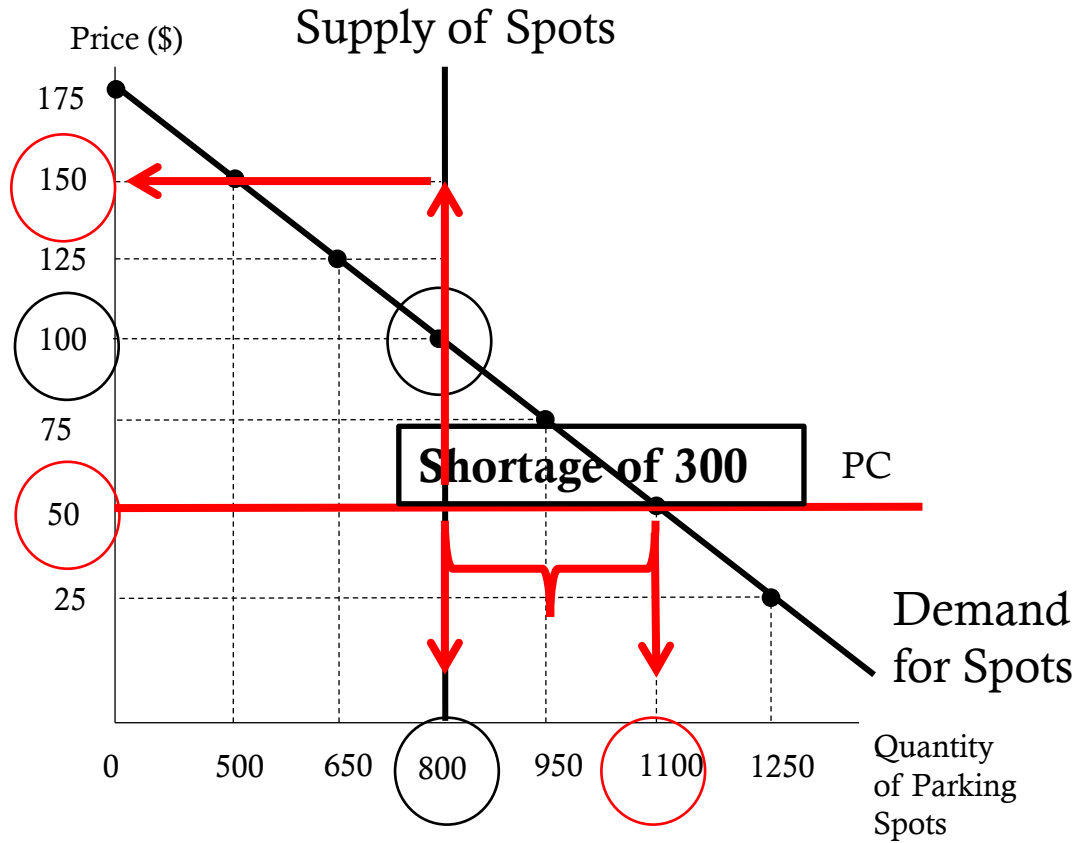
- Surplus = 50,000 workers (unemployment).
- Some workers benefit from higher pay, but others lose jobs or hours.
- Firms may automate or shift toward part-time/contract labor.
- Long-run adjustment reduces low-skill employment opportunities.

Minimum Wage



- Let's examine the labor market.
- Those who want to work are represented by the supply of labor.
- The firms who want to use labor are represented by the demand for labor.
- The market equilibrium wage is \$5.25 and 30,000 workers are employed.
- The government believes this wage to be unfair to workers and imposes a minimum wage (price floor) of \$7.25.
- Labor demand falls to 10,000. Labor supply increases to 50,000.
- A surplus of labor (unemployment) of 40,000 is created.

University Parking



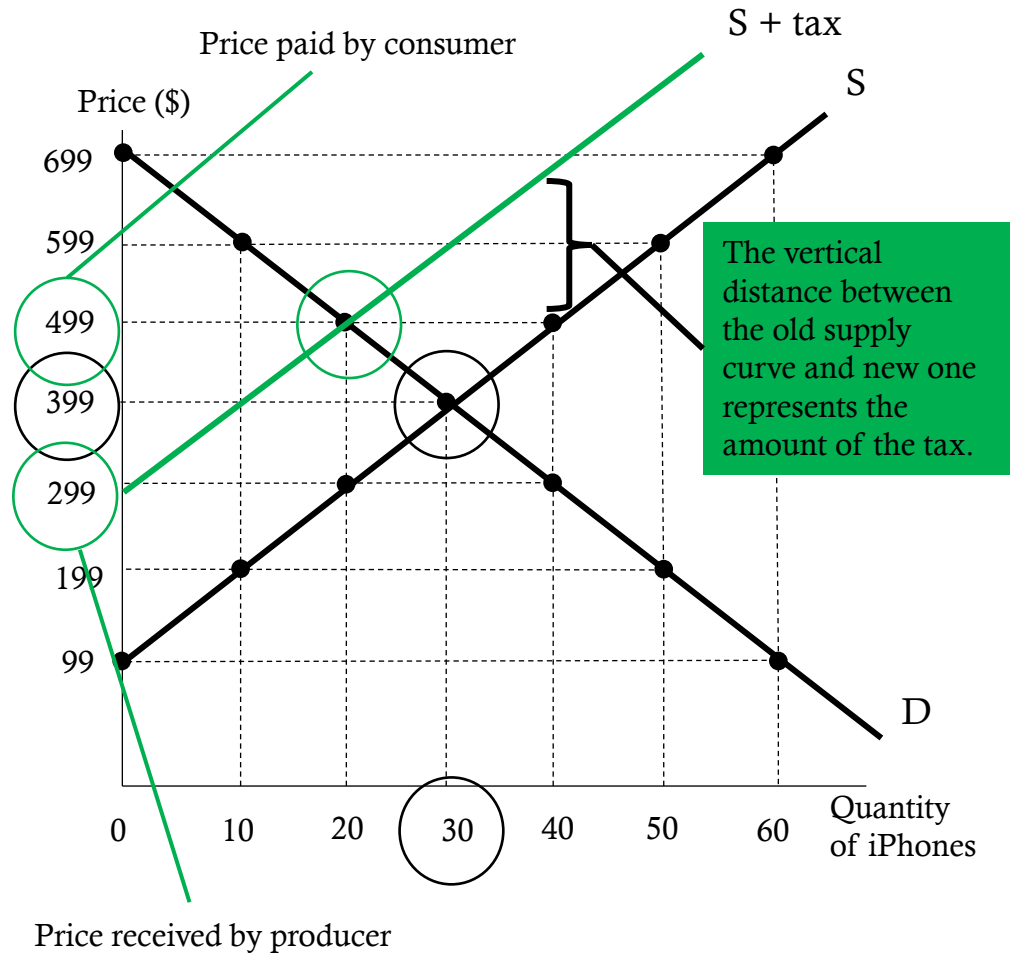
Implicitly Subsidizing Rent

- Let's look at a common public policy designed to help those obtain a place to live.
- First, consider people who are in arrears (owing money that should have been paid earlier).
- Landlords may seek to raise rent on said people as they seem to be more likely to default and landlords want to get new tenants in who are more agreeable to pay rent on time.
- However, policies are put in place to protect these renters from rent hikes.
- The end result is a housing imbalance. First, landlords become more selective in their leasing (often excluding young people and those on fixed-term employment). Second, with protection in rent, housing becomes more restricted and of poorer quality (our rent control model).

Taxes

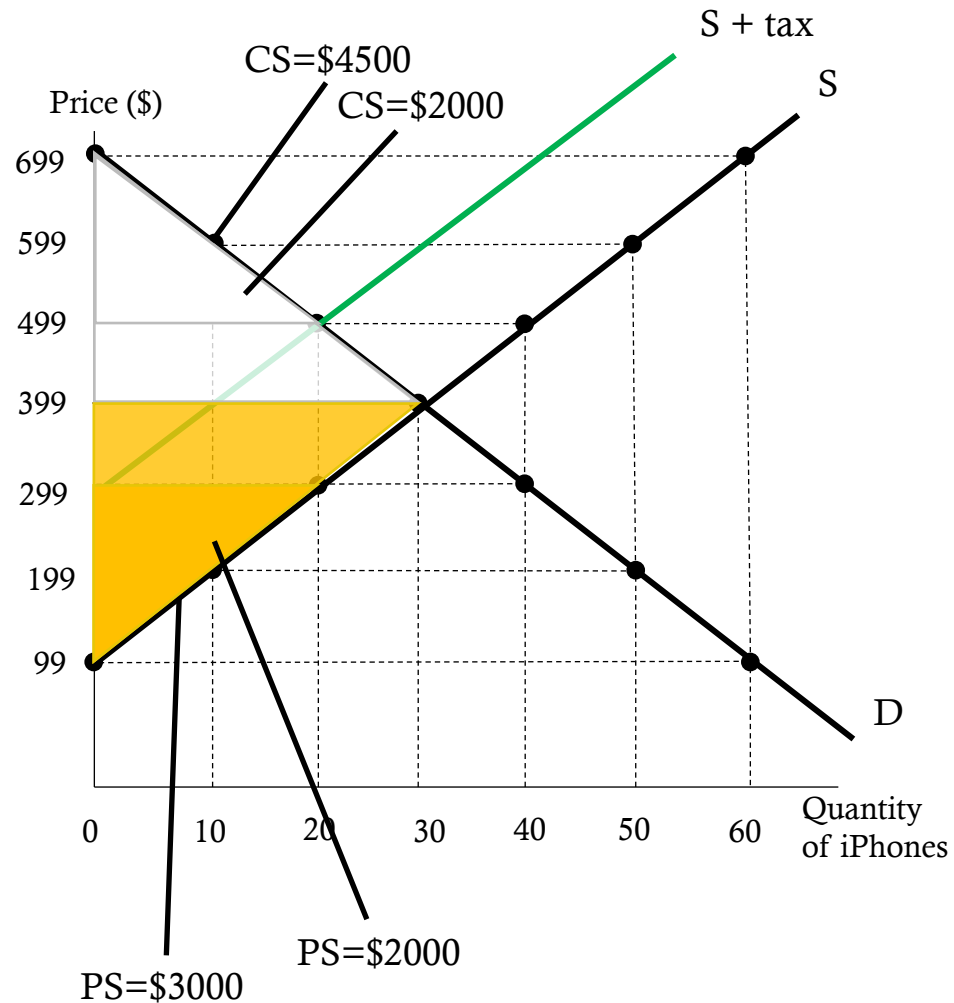
- **Recall:** one of the determinants of supply is a change in taxes.
- An increase in taxes causes a **decrease** in supply due to the increase in the cost of production per unit.
- A decrease in taxes causes an **increase** in supply due to the decrease in the cost of production per unit.
- Taxes are set by the government and usually result in the consumer paying a higher price and receiving a lower quantity than the market equilibrium. The producer will also receive a lower price and produce less than the market equilibrium.
- As a result of the market being distorted by taxes, consumer and producer surplus decline and welfare is lost to society.

Taxes



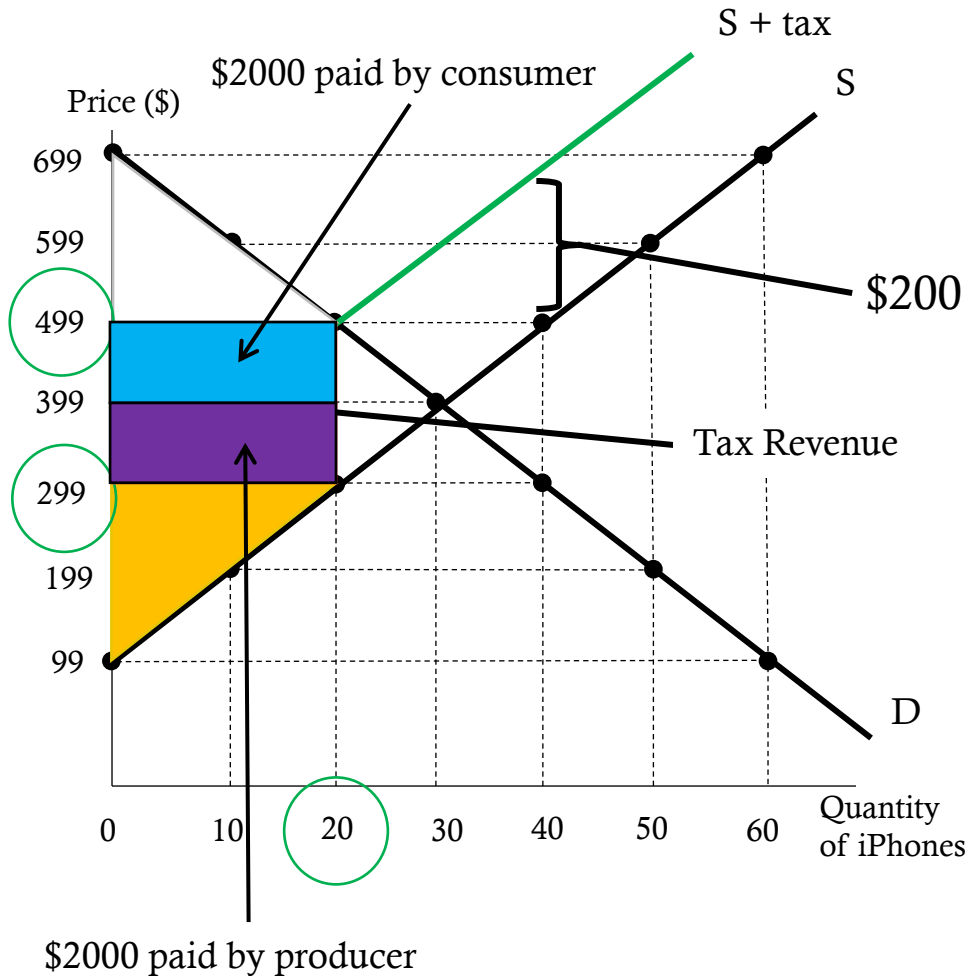
- Let's reconsider the market for iPhones.
- Market equilibrium is established at a price of \$399 and 30 iPhones.
- Suppose the government imposes a tax of \$200 per unit sold.
- The result is a **decrease** in supply to S + tax.
- Notice the new equilibrium is at a price of \$499 and 20 iPhones. However, the tax was \$200.
- Why isn't the new equilibrium at \$599 and 10 iPhones?
- Because some of the tax (\$100) is paid by the consumer and some (\$100) paid by the producer.

Tax Incidence



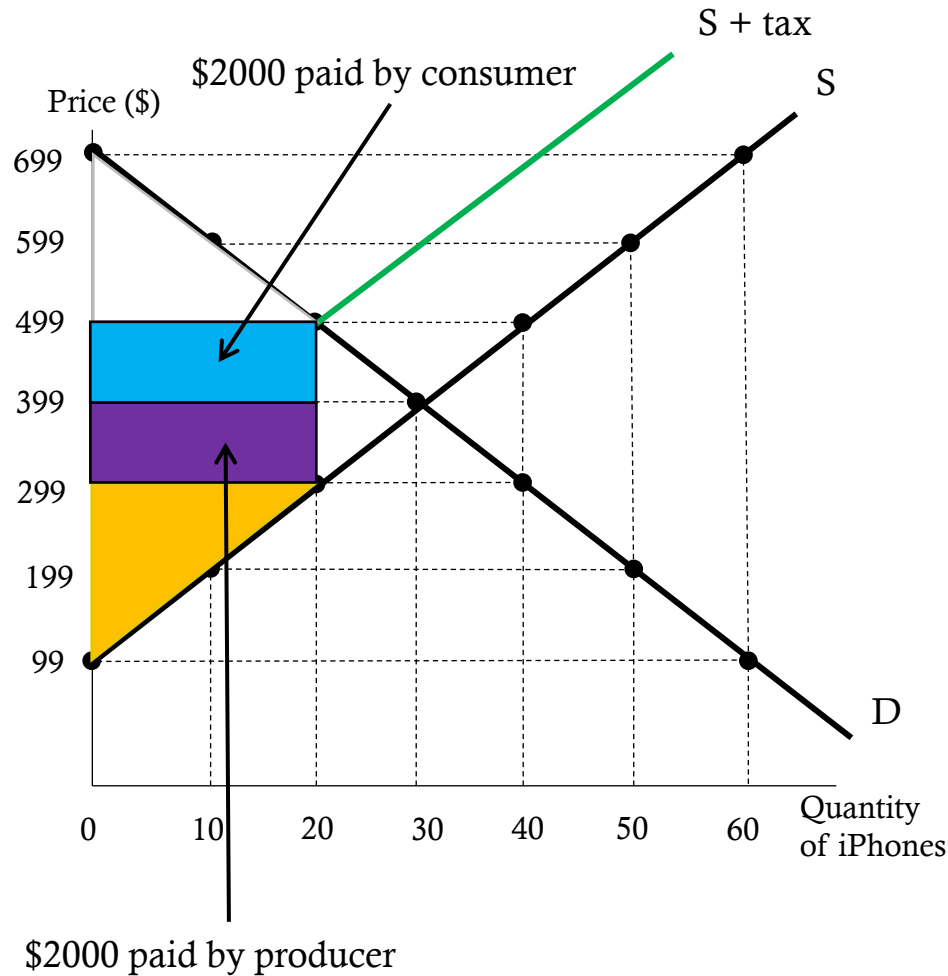
- The market equilibrium of \$399 and 30 iPhones gives us a consumer surplus of
 $CS: \frac{1}{2}((30-0)*(699-399)) = \4500
or
 $CS: \frac{1}{2}(30*300) = \4500
- and a producer surplus of
 $PS: \frac{1}{2}((30-0)*(399-99)) = \4500
or
 $PS: \frac{1}{2}(30*300) = \4500 .
- Let's impose a \$200 tax. Our new supply curve will be S + tax.
- The new equilibrium is \$499 and 20 iPhones. The consumer pays \$499, and the producer receives \$299.
- This limits our consumer and producer surplus to:
 - $CS: \frac{1}{2}(20*200) = \2000
 - $PS: \frac{1}{2}(20*200) = \2000
- By introducing a tax, consumer and producer surplus both fall for a total surplus loss of
- $\$9000 - \$4000 = \$5000$

Tax Incidence



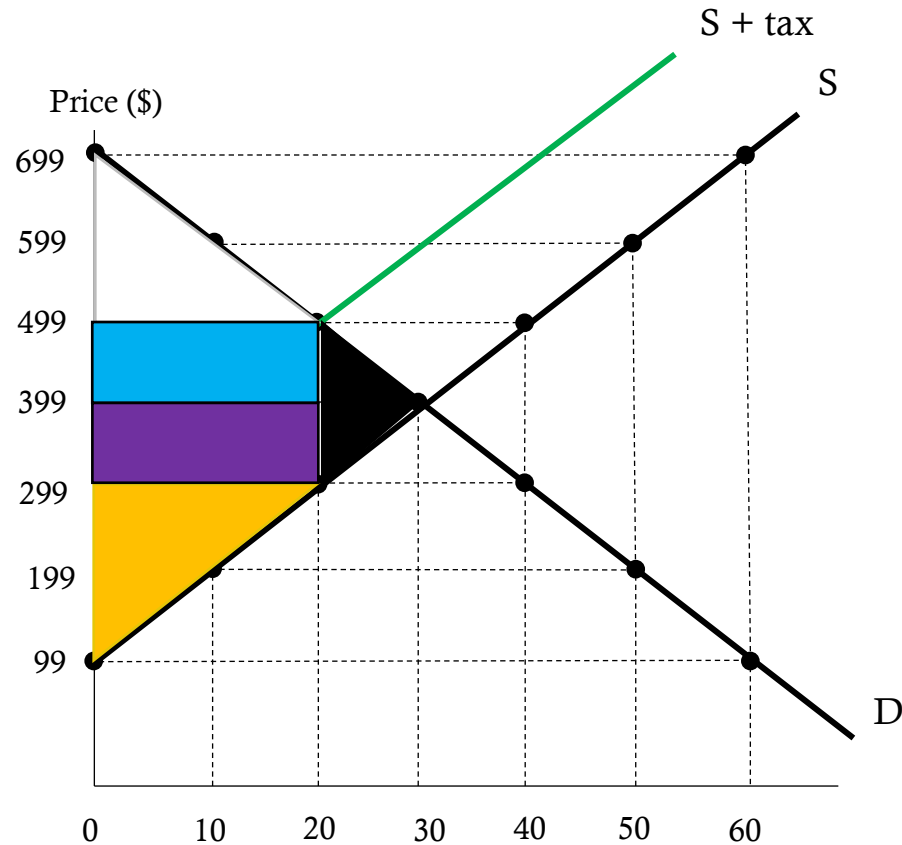
- So far, we have seen how consumer and producer surplus are affected. What about the tax?
- **Recall:** the vertical distance between the two supply curves is the amount of the tax.
- At the new equilibrium, consumers pay \$499, and producers receive \$299 and the quantity bought and sold is 20.
- Since taxes generate revenue for the government, then the government collects the tax times the quantity sold ($t \cdot Q$)
- In our case, tax revenue is: $TR: \$200 \cdot 20 = \4000
- To find how much the consumer and producer contribute towards the tax simply subtract the price they pay/receive from the original equilibrium price of \$399

Tax Incidence



- We have seen how much the consumer and producer contribute towards the tax and tax revenue.
- **Tax incidence** – the analysis of how much of the tax burden the consumer and producer each bear.
- In our graph, the consumer bears \$100 of the tax and the producer bears the other \$100. They each pay an equal amount of tax revenue (\$2000).
- This means that, in our market, the tax incidence is evenly split.

Deadweight Loss



- We have nearly accounted for the original consumer and producer surplus areas.
- We have a small triangle left in our graph.
- This triangle represents the deadweight loss due to taxation.
- It arises because consumers and producers would have preferred the original equilibrium of \$399 and 30 iPhones.

Activity – The Soda Tax Debate

- The government imposes a **\$0.50 per can** soda tax.
- Before tax: price = \$1.50, quantity = 1 million cans.
- After tax: consumers pay **\$1.70**, producers receive **\$1.20**, quantity sold = 800,000 cans.
- How is the tax burden shared?
- How much tax revenue is collected?
- Identify the deadweight loss area conceptually.

Activity – The Soda Tax Debate

- Consumer burden = $\$0.20 \times 800,000 = \$160,000$
- Producer burden = $\$0.30 \times 800,000 = \$240,000$
- Total tax revenue = **\$400,000**
- Deadweight loss = lost transactions between 800,000 and 1,000,000 units.
- Elasticities determine who bears more burden (side that's less elastic pays more).

Key Takeaways

- **Price controls and taxes distort market equilibrium.**
Price ceilings and floors prevent prices from performing their natural rationing role.
- **Price ceilings (like rent control) create shortages**, misallocation, and black markets.
Although intended to help consumers, they often reduce availability and quality.
- **Price floors (like minimum wages) create surpluses**, often in the form of unemployment or unsold goods.
- **Taxes create wedges** between the prices buyers pay and sellers receive, reducing the quantity traded.
The burden is shared depending on elasticity—whichever side is **less elastic bears more** of the tax.
- **Deadweight loss** measures the efficiency cost of intervention—output that would have occurred in an unregulated market.

Knowledge Check

1. A price ceiling set below the equilibrium price causes:
2. A binding price floor leads to:
3. A tax on sellers shift the:
4. The area representing lost gains from trade due to taxes or price controls is called:
5. A minimum wage above equilibrium causes:

Knowledge Check Answers

1. A shortage
2. Excess supply (surplus)
3. Supply curve upward (or left)
4. Deadweight loss
5. Labor surplus (unemployment)